

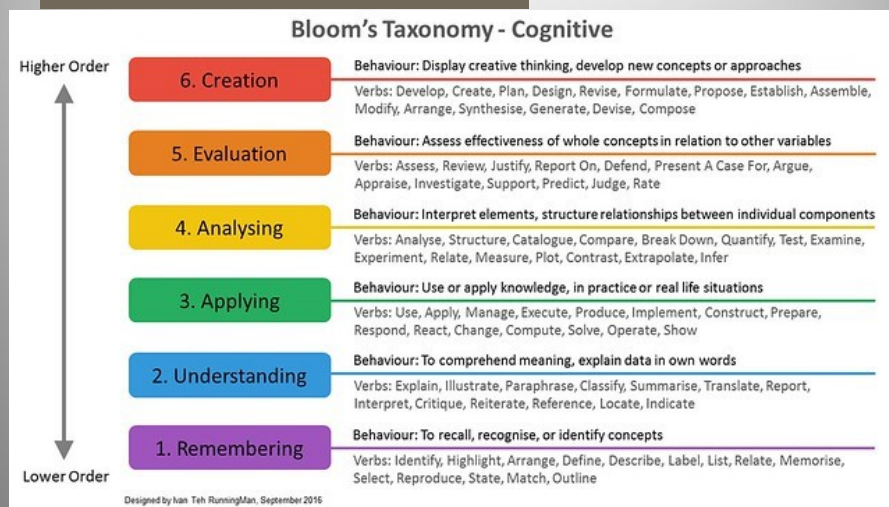
CS 322: Software Construction

□ Course Teacher: Dr. Muddassar Azam Sindhu

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Bloom's Taxonomy



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CLOs

CS-322 Software Construction

S. No.	CLO	BT Level
CLO1	Explain the concepts & activities in the entire software engineering process focusing on construction and major umbrella activities	C2
CLO2	Develop work products for software development activities	C3
CLO3	Develop a software system according to the developed analysis and design models for a given set of requirements, test and evaluate it	C5
CLO4	Analyze a design/compare designs for good design principles	C4

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Weekly Lecture Plan

Week	Lecture / Lab
1	Introduction to project planning and phases, Introduction to projects and allocation, Project Plan Development
2	Requirements / Specifications, Static Checking, Acceptance Tests, Code Review
3	Version Control, Specifications, Designing Specifications
4	Development of analysis models
5	SRS review and refinement
6	Design principles, data design
7	Interface design
8	Detailed design
9	Review of design and refinement
10	Coding, Introduction to Java, Building a front-end with Java
11	Refining the front-end
12	Database connectivity in Java
13	Refinement of classes and traceability with design
14	Testing of the software system using the Junit Tool
15	Project Presentations
16	Project Presentations

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Evaluation Criteria

- ☐ Final Term: 50 Marks
- ☐ Mid Term: 10 Marks
- ☐ Planning: 5 Marks
- ☐ Analysis: 7 Marks
- ☐ Design: 8 Marks
- ☐ Implementation: 10 Marks
- ☐ Testing: 5 Marks
- ☐ Quizzes: 5

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Recommended Readings

- ☐ Pratap Mohapatra, Software Engineering A Lifecycle Approach, New Age Publications, 2010
- ☐ Argo UML User Manual (Available Online)
- ☐ Craig Larman, Applying UML and Patterns, Pearson Education, 3rd Edition, 2004
- ☐ Ian Sommerville, Software Engineering, Addison Wesley, 3rd Edition, 2010
- ☐ Roger S. Pressman, Software Engineering - A Practitioner's Approach, McGraw Hill, 7th Edition, 2009
- ☐ John Lewis & William Loftus, Java Software Solutions, Pearson Education, 4th Edition, 2005

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Tools / IDEs

- ☐ Java SE 17 with IDEs (Eclipse/NetBeans) and Command Prompt / Notepad
- ☐ Project Libre (For Project Management Tasks)
- ☐ Argo UML / Star UML (For UML Modelling)
- ☐ JUnit for Software Testing

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Course Logistics

- ☐ Class Groups to be formed and reported to the instructor before the next lab session on Thursday.
- ☐ Preferable to have Java/Project Libre / Argo UML installed on your own laptops
- ☐ Piazza / Google Group for the course
- ☐ Projects to be assigned to each group before the first lab session
- ☐ Contact Hours: Thursdays (0930-1130)
 - You can also drop me an email on masindhu@qau.edu.pk if I am not visible in the department during office hours even though I am here most of the time. ☺

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Software Project Management

Week 1

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Project Management

☐ Involves the

- Planning
- Monitoring
- Control

of people, process and events that occur as a software evolves from a concept to an operational implementation

☐ Work Product

- Project Plan

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The 4 P's

- ❑ People — stakeholders
- ❑ Product — the software to be built
- ❑ Process — the set of framework activities and software engineering tasks to get the job done
- ❑ Project — all work required to make the product a reality

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Planning Practices

- ❑ Principles
 - Understand the project scope
 - Involve the customer (and other stakeholders)
 - Estimate based on what you know
 - Consider risk
 - Be realistic
 - Adjust granularity as you plan
 - Define how quality will be achieved
 - Define how you'll accommodate changes
 - Track what you've planned

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Project Planning Task Set

- ☐ Establish project scope
- ☐ Determine feasibility
- ☐ Analyze risks
- ☐ Define required resources
 - Determine required human resources
 - Define reusable software resources
 - Identify environmental resources

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Project Planning Task Set-II

- ☐ Estimate cost and effort
 - Decompose the problem
 - Develop two or more estimates
 - Reconcile the estimates
- ☐ Develop a project schedule
 - ☐ Establish a meaningful task set
 - ☐ Define a task network
 - ☐ Use scheduling tools to develop a timeline chart
 - ☐ Define schedule tracking mechanisms

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What is Scope?

- ❑ *Software scope* describes
 - the functions and features that are to be delivered to end-users
 - the data that are input and output
 - the performance, constraints, interfaces, and reliability that *bound* the system.
- ❑ Scope is defined using one of two techniques:
 - ❑ A narrative description of software scope is developed after communication with all stakeholders.
 - ❑ A set of use-cases is developed by end-users.

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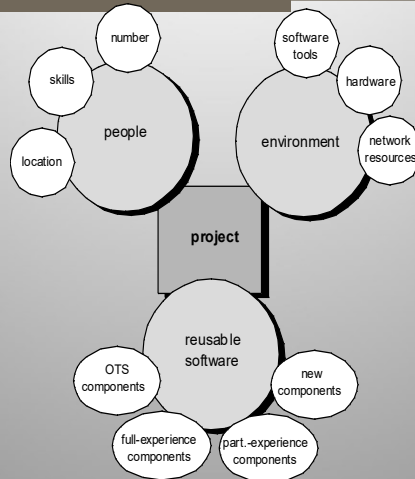
Estimation

- ❑ Estimation of resources, cost, and schedule for a software engineering effort requires
 - experience
 - access to good historical information (metrics)
 - the courage to commit to quantitative predictions when qualitative information is all that exists
- ❑ Estimation carries inherent risk and this risk leads to uncertainty

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Resources



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Project Estimation



- ☐ Project scope must be understood
- ☐ Elaboration (decomposition) is necessary
- ☐ Historical metrics are very helpful
- ☐ At least two different techniques should be used
- ☐ Uncertainty is inherent in the process

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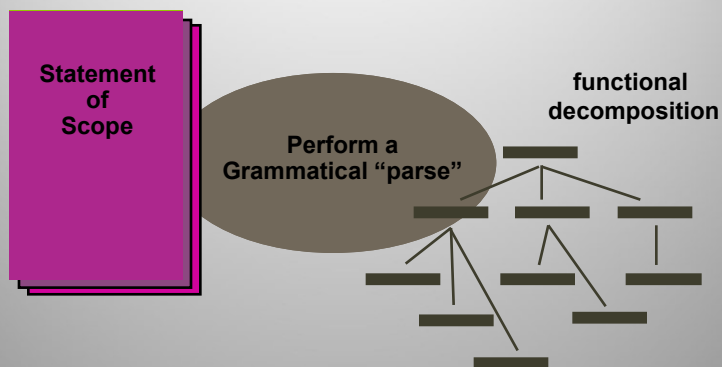
Estimation Techniques

- ☐ Past (similar) project experience
- ☐ Conventional estimation techniques
- ☐ Empirical models
- ☐ Automated tools

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Functional Decomposition



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LOC Based Estimation

Function	Estimated LOC
User interface and control facilities (UICF)	2,300
Two-dimensional geometric analysis (2DGA)	5,300
Three-dimensional geometric analysis (3DGA)	6,800
Database management (DBM)	3,350
Computer graphics display facilities (CGDF)	4,950
Peripheral control function (PCF)	2,100
Design analysis modules (DAM)	8,400
<i>Estimated lines of code</i>	33,200

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Process-Based Estimation Example

Activity	CC	Planning	Risk Analysis	Engineering		Construction Release		CE	Totals
Task				analysis	design	code	test		
Function									
UICF				0.50	2.50	0.40	5.00	n/a	8.40
2DGA				0.75	4.00	0.60	2.00	n/a	7.35
3DGA				0.50	4.00	1.00	3.00	n/a	8.50
CGDF				0.50	3.00	1.00	1.50	n/a	6.00
DSM				0.50	3.00	0.75	1.50	n/a	5.75
PCF				0.25	2.00	0.50	1.50	n/a	4.25
DAM				0.50	2.00	0.50	2.00	n/a	5.00
Totals	0.25	0.25	0.25	3.50	20.50	4.50	16.50		46.00
% effort	1%	1%	1%	8%	45%	10%	36%		

CC = customer communication CE = customer evaluation

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Scheduling Principles

- ❑ compartmentalization—define distinct tasks
- ❑ interdependency—indicate task interrelationship
- ❑ effort validation—be sure resources are available
- ❑ Time allocation—Number of work units, start/end date
- ❑ defined responsibilities—people must be assigned
- ❑ defined outcomes—each task must have an output
- ❑ defined milestones—review for quality

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Task Set Refinement

1.1 Concept scoping determines the overall scope of the project.

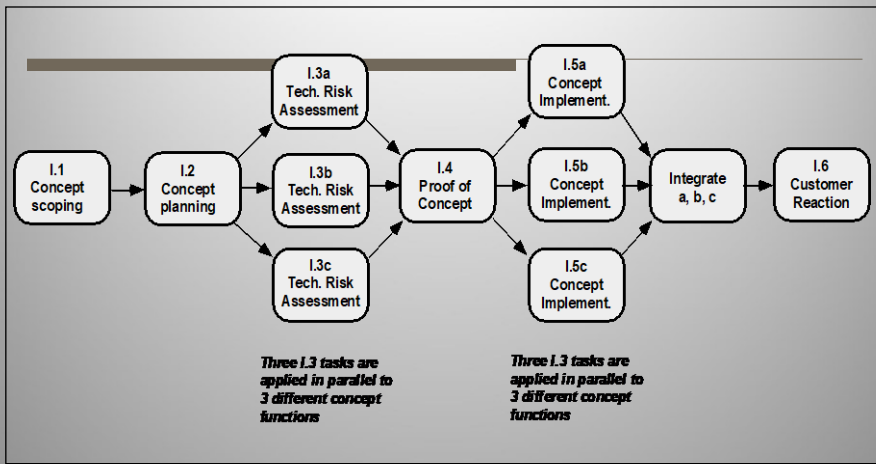


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Task definition: Task 1.1 Concept Scoping
1.1.1 Identify need, benefits and potential customers;
1.1.2 Define desired output/control and input events that drive the application;
Begin Task 1.1.2
1.1.2.1 FTR: Review written description of need
FTR indicates that a formal technical review (Chapter 26) is to be conducted.
1.1.2.2 Derive a list of customer visible outputs/inputs
1.1.2.3 FTR: Review outputs/inputs with customer and revise as required;
endtask Task 1.1.2
1.1.3 Define the functionality/behavior for each major function;
Begin Task 1.1.3
1.1.3.1 FTR: Review output and input data objects derived in task 1.1.2;
1.1.3.2 Derive a model of functions/behaviors;
1.1.3.3 FTR: Review functions/behaviors with customer and revise as required;
endtask Task 1.1.3
1.1.4 Isolate those elements of the technology to be implemented in software;
1.1.5 Research availability of existing software;
1.1.6 Define technical feasibility;
1.1.7 Make quick estimate of size;
1.1.8 Create a Scope Definition;
endTask definition: Task 1.1
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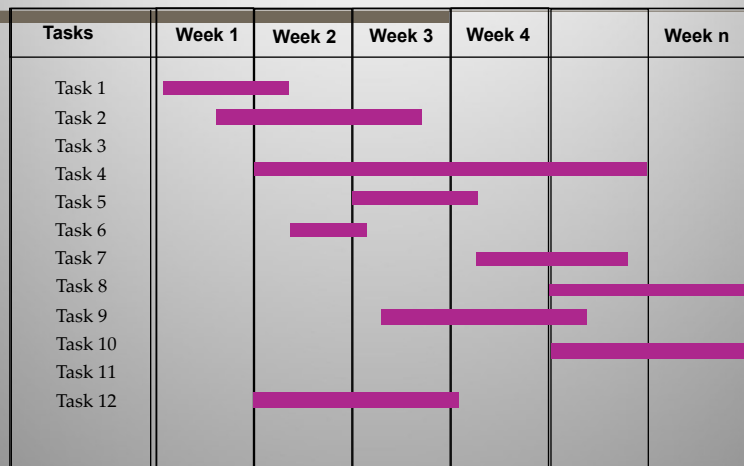
Define a Task Network



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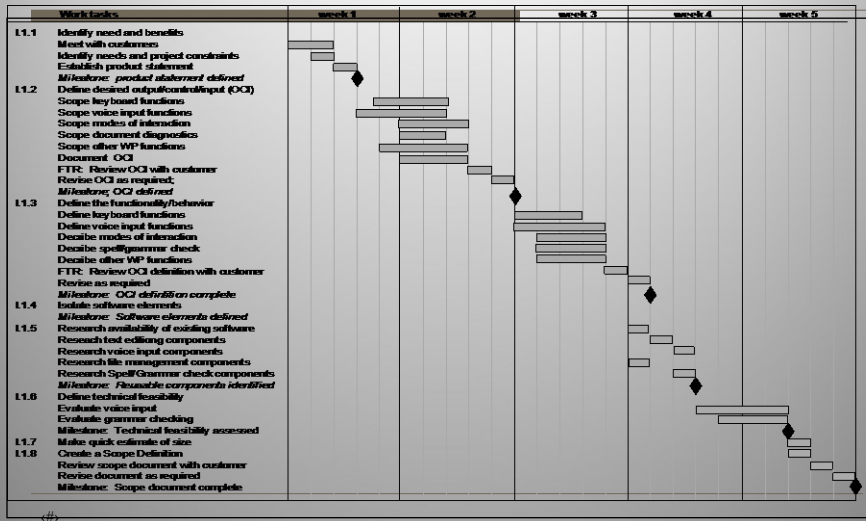
Timeline Charts



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Use Automated Tools to Derive a Timeline Chart



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Schedule Tracking

- conduct periodic project status meetings in which each team member reports progress and problems.
- evaluate the results of all reviews conducted throughout the software engineering process.
- determine whether formal project milestones have been accomplished by the scheduled date.
- compare actual start-date to planned start-date for each project task listed in the resource table
- meet informally with practitioners to obtain their subjective assessment of progress to date and problems on the horizon.

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